



*...Or how to quickly harness
the potential of LLMs*





Jean-François (Fran) Gutierrez ✓ (He/Him)

AI Augmented Software Engineering Advocate | 🤖 A.K.A. "The Go-To Guy for Gen AI Insights"

Valencia, Valencian Community, Spain · [Contact info](#)



🤖 Tailor-Made AI Solutions

💻 Software Consultancy

🎤 Tech Events

🔗 Google Developer Groups



Formación



/GutierrezFrancois



GitHub

/DrZuzzjen

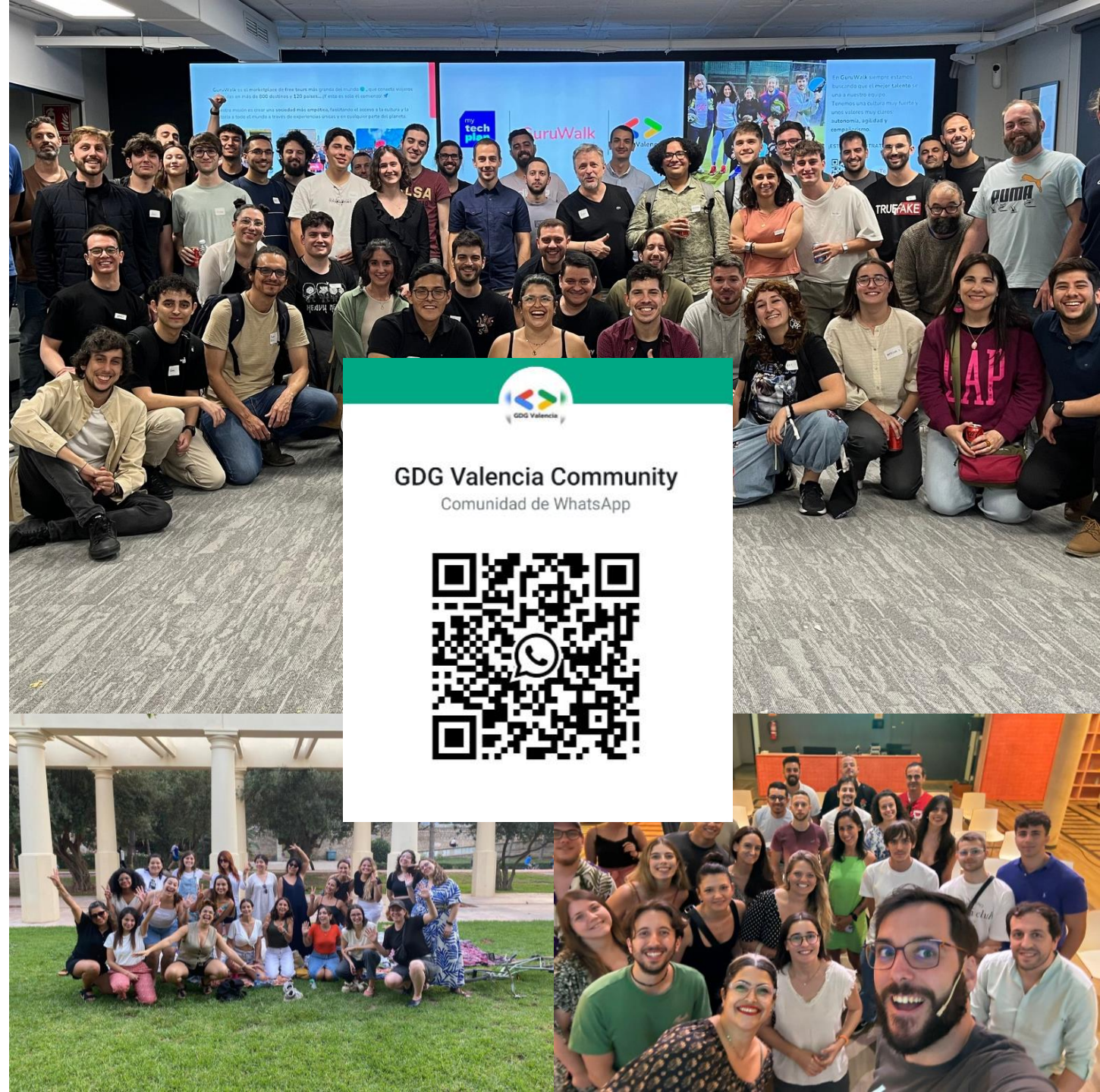


/DrZuzzjen



HUGGING FACE
/FranZuzz







Use case

Simple chatbot that can use
multiple agents connected to internal API's

Let's pick our tools

LANGGRAPH

CREWAI

SWARM



Semantic Kernel

LangGraph



LangChain



Microsoft
AutoGen

crewai

Agentic Frameworks: The Complete Guide to the Systems Used in Building Autonomous Agents

Amy Brennan, Content Marketing Manager

Medium

Search

Top 5 Agentic AI Frameworks to Watch in 2025



Lekha Priya · [Follow](#)

4 min read · Jan 9, 2025



r/LLMDevs · 4 mo. ago

Best framework/reading list for agentic systems?

Discussion

"Agentic AI" seems to be a fairly recent buzzword, and as far as I know is the most well-known term for in "self-driving" AI agents (now that the term "ai agent" has essentially become a synonym for GPTs, which are pre-prompted chat bots that are primarily triggered by direct human input).

Essentially, LLMs talking to each other to reach a goal, and then fulfilling that goal by interacting with the world.

Blog

Tutorials

Case Studies

Company

Eng

Best 5 Frameworks To Build Multi-Agent AI Applications

What are the Top Agentic Frameworks? A guide to AI Agents



Erich Hellstrom
Oct 31, 2024 — 6 min read

PromptLayer

IBM

Hybrid Cloud

Products

Consulting

Support

Think 2025

Think

Artificial intelligence

Cloud

Security

Videos

Reports

Podcasts

Events

More

AI agent frameworks: Choosing the right foundation for your business



eworks



Which one?

<> Code Issues 324 Pull requests 44 Discussions Actions Projects 2 Security Insights

langchain Public Watch 738 Fork 17k Starred 105k

master 882 Branches 851 Tags Go to file Add file <> Code About

<> Code Issues 477 Pull requests 38 Discussions Actions Projects Security Insights

autogen Public Watch 451 Fork 6.4k Starred 42.6k

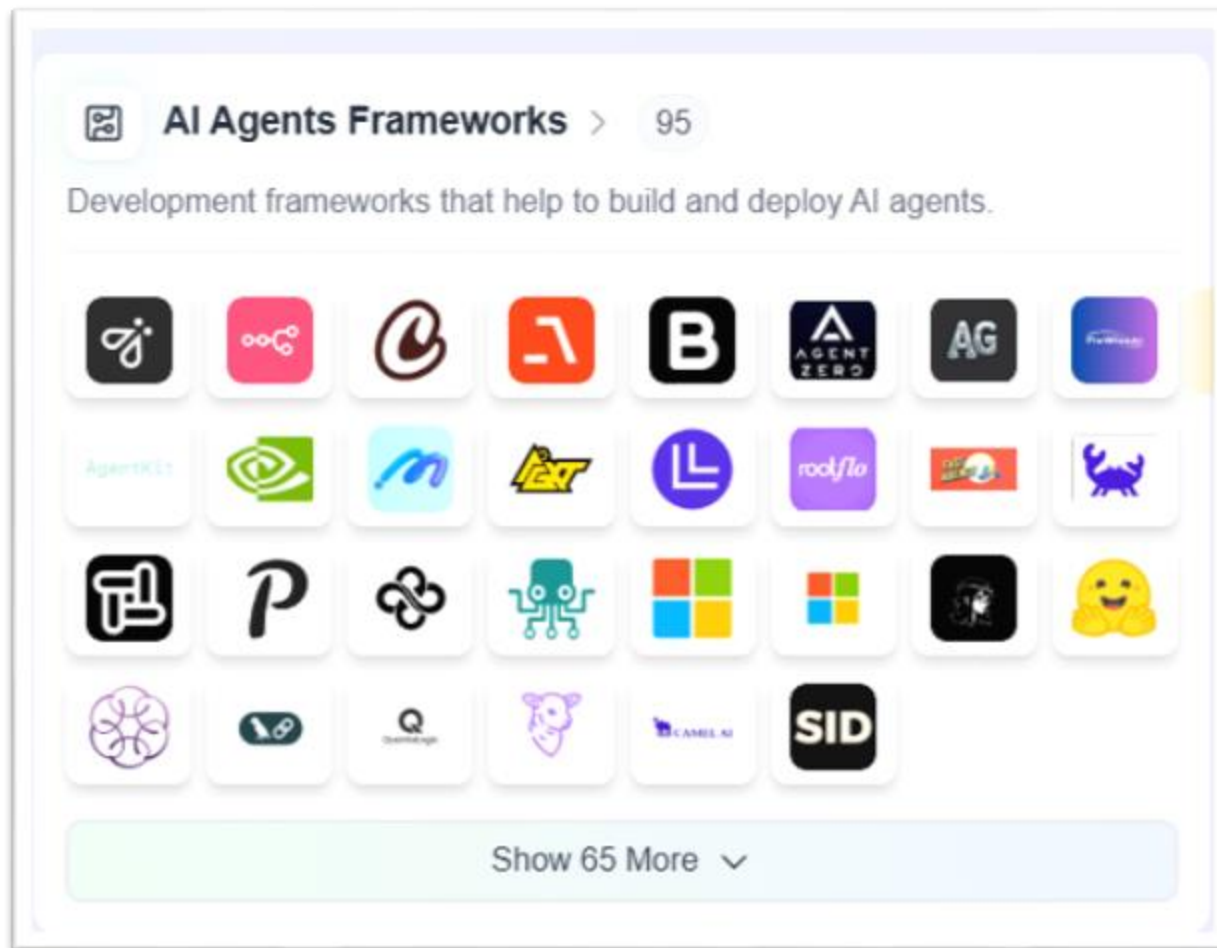
main 187 Branches 108 Tags Go to file Add file <> Code About

<> Code Issues 79 Pull requests 36 Discussions Actions Projects Security Insights

crewAI Public Watch 280 Fork 4k Starred 29.4k

main 381 Branches 79 Tags Go to file Add file <> Code About

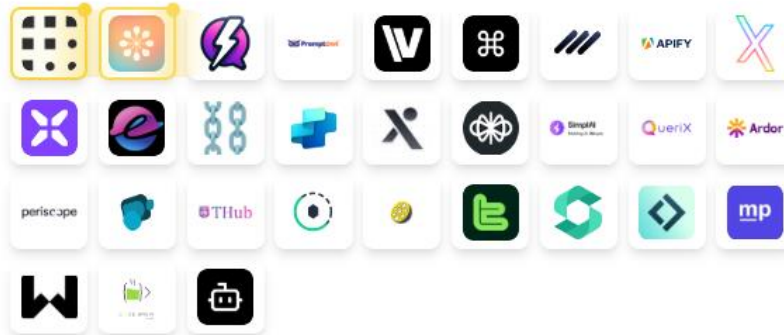
AI Agents Market Landscape





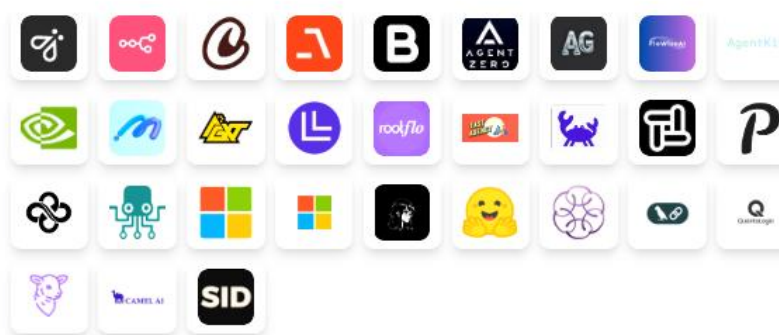
AI Agents Platform > 133

Platforms that allow users to create, customize, and deploy AI agents.

[Show 103 More](#)

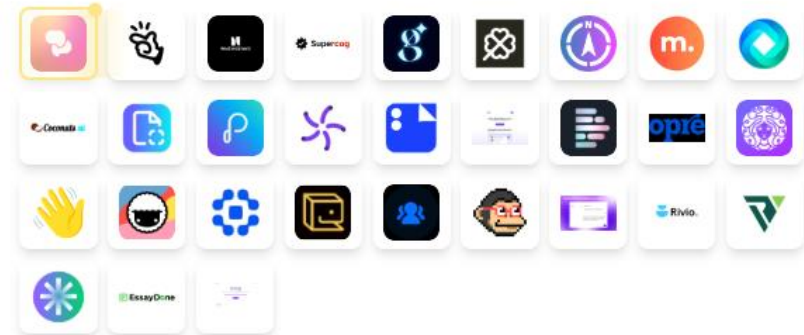
AI Agents Frameworks > 95

Development frameworks that help to build and deploy AI agents.

[Show 65 More](#)

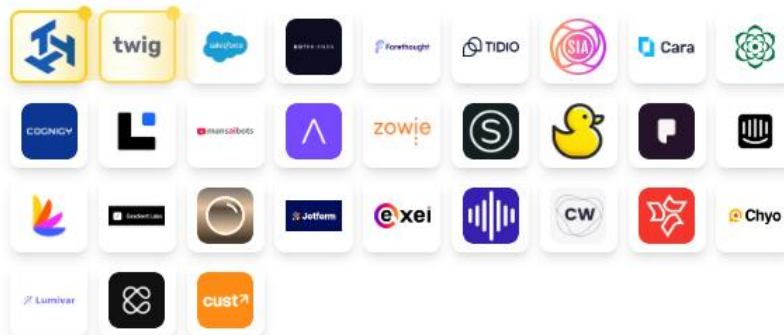
Productivity > 91

AI assistants that help you get more done by automating repetitive tasks.

[Show 61 More](#)

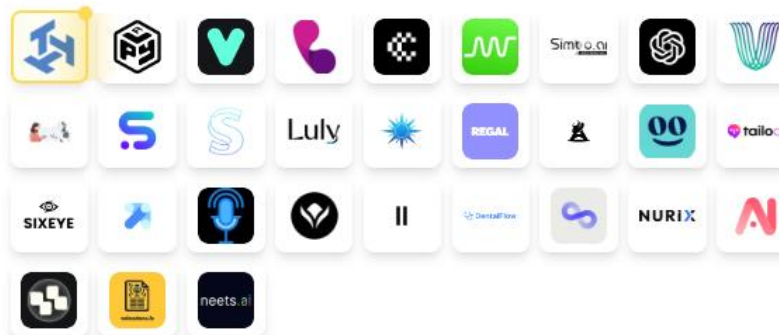
Customer Service > 64

Virtual assistants that handle customer inquiries, support tickets, and service...

[Show 34 More](#)

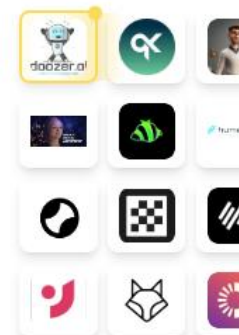
Voice AI Agents > 60

AI assistants that can understand and respond to voice commands and conduct...

[Show 30 More](#)

Digital Workers > 52

AI agents that function as:



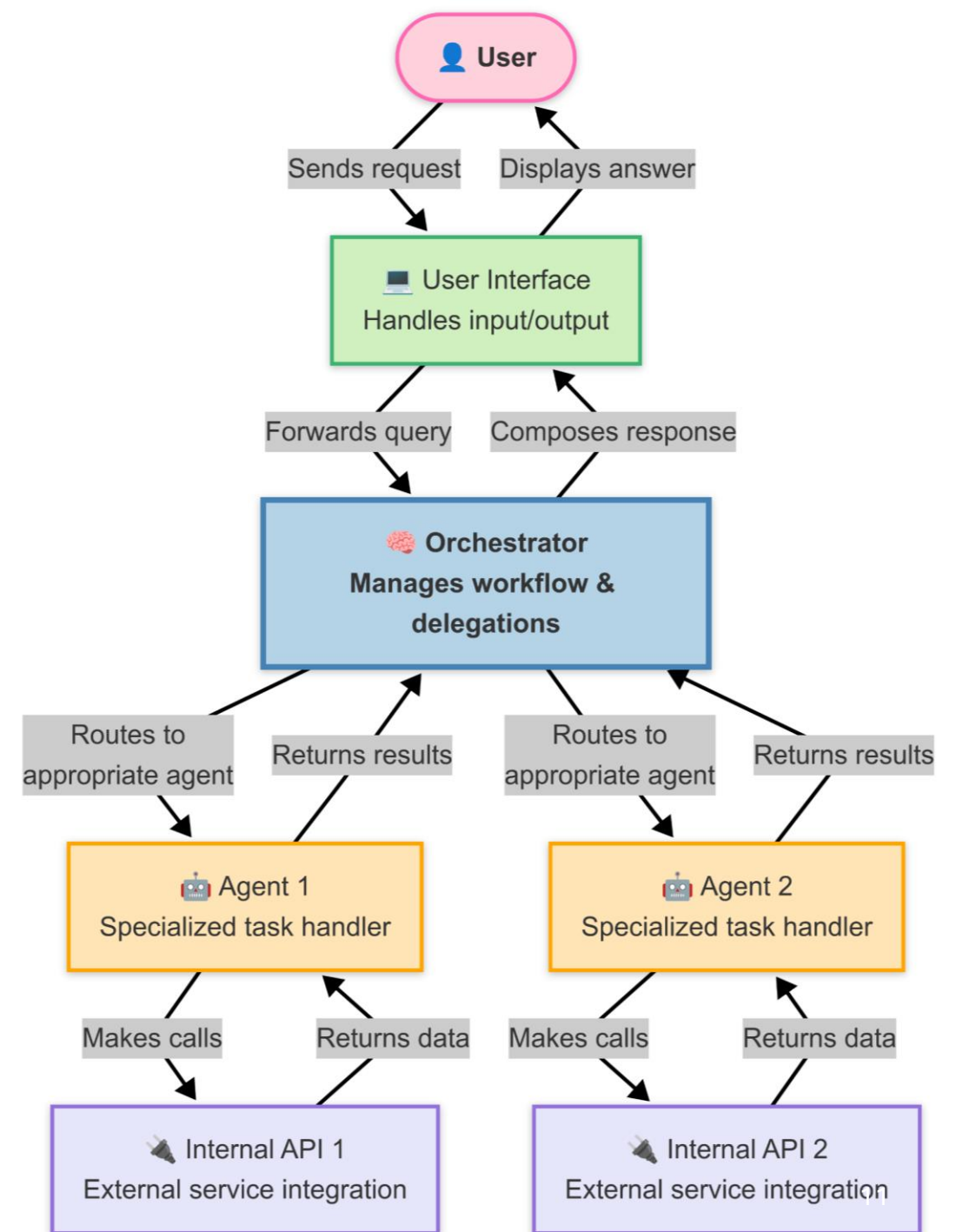
Chatbot + API's

- Langgraph Orchestrator
- Langchain/tools for API's
- Simple streamlit app for UI



LangGraph

4/15/2025



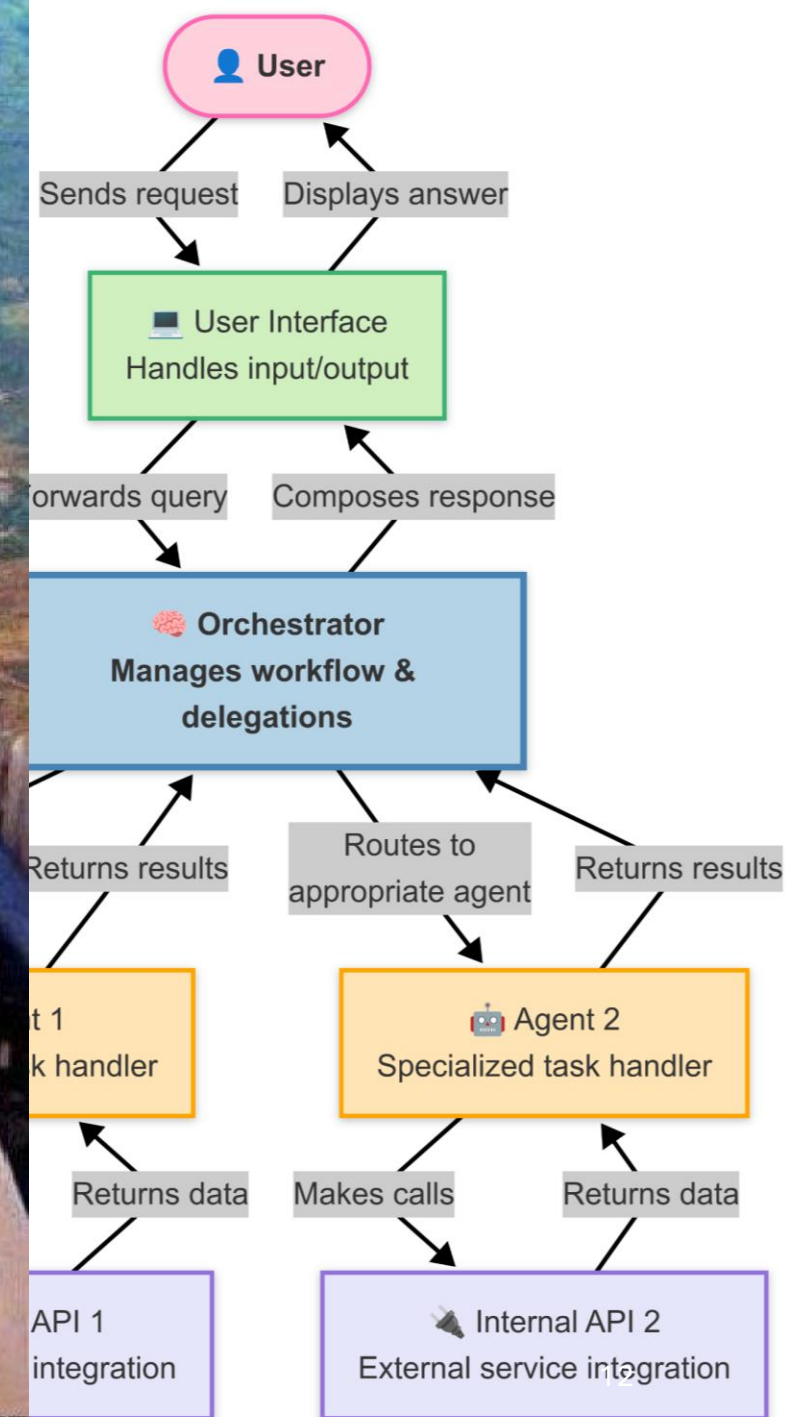
Chatbot + API

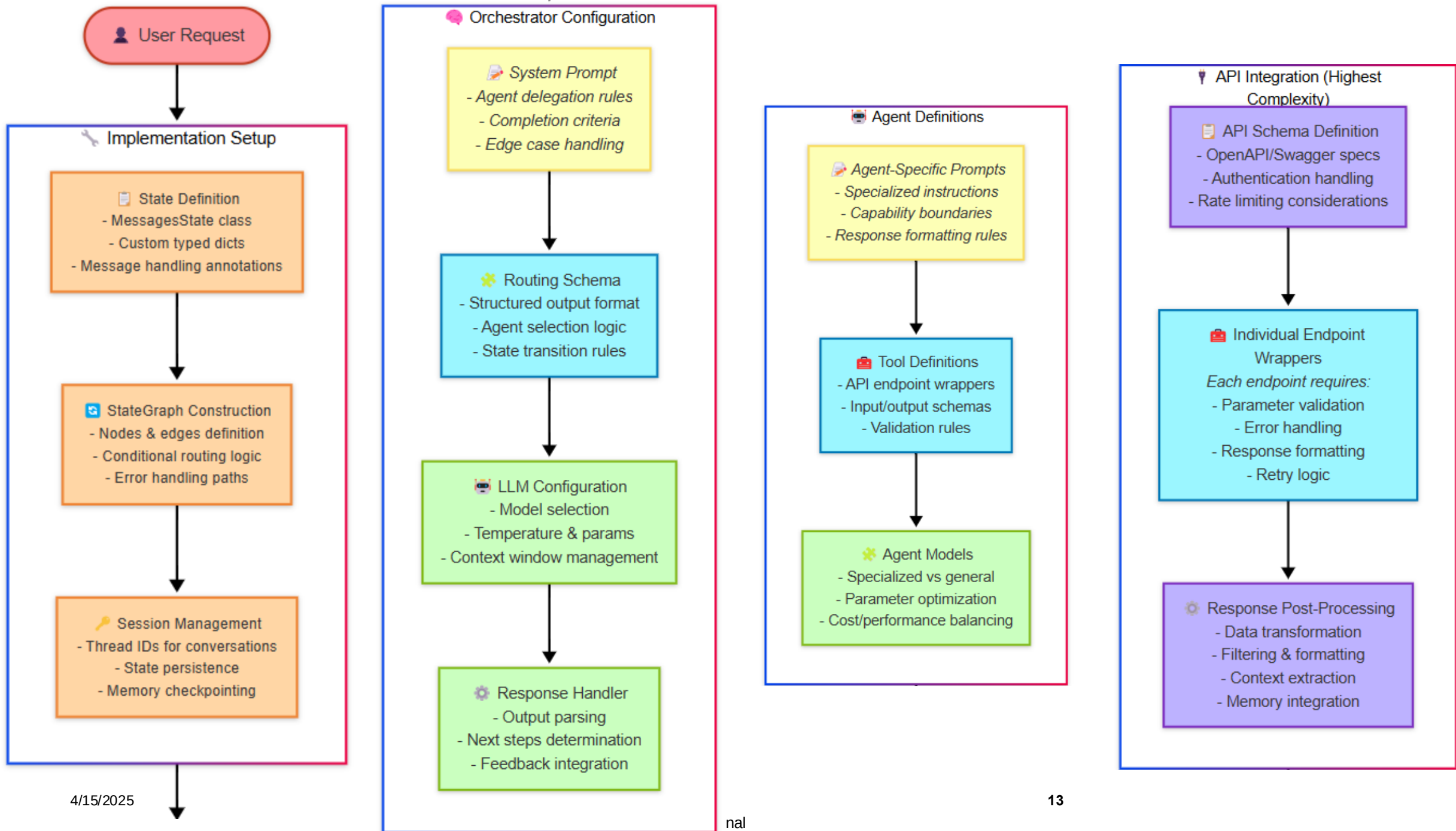
- Langgraph Orchestrator
- Langchain/tools for API's
- Simple streamlit app for UI

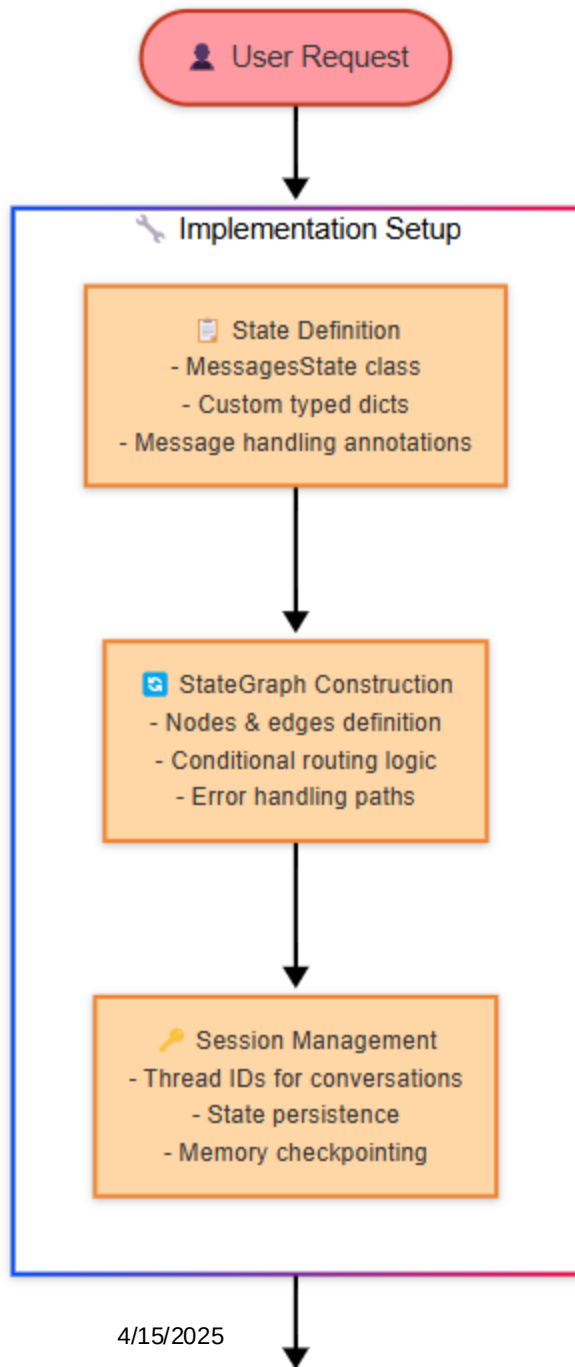


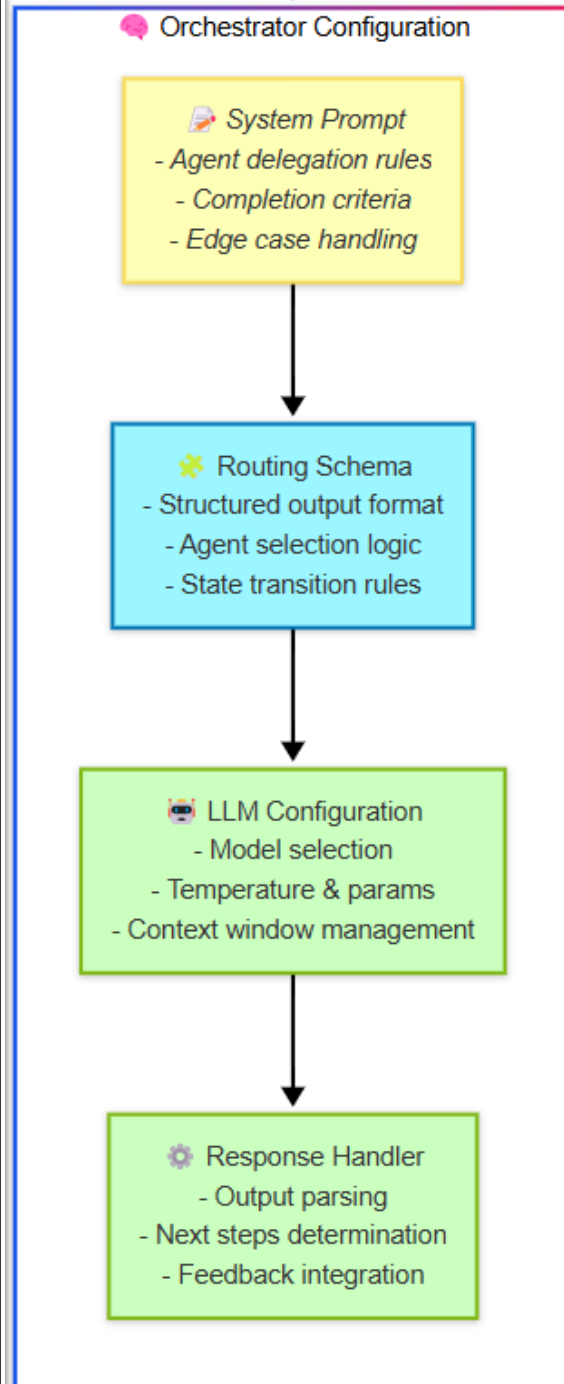
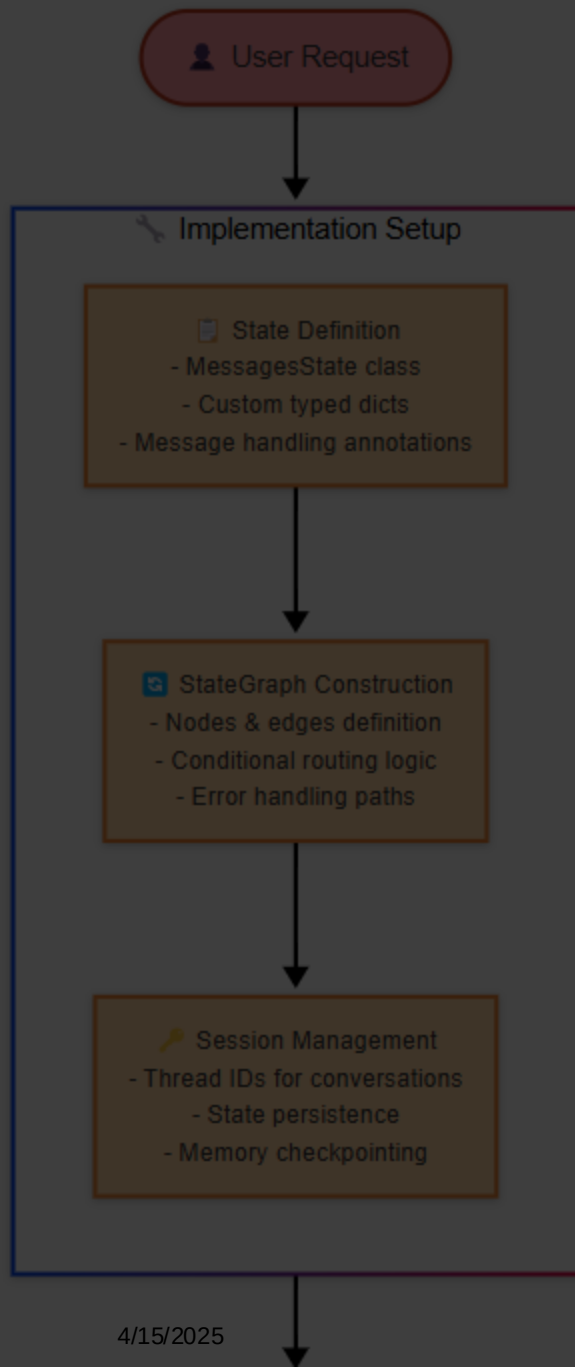
LangGraph

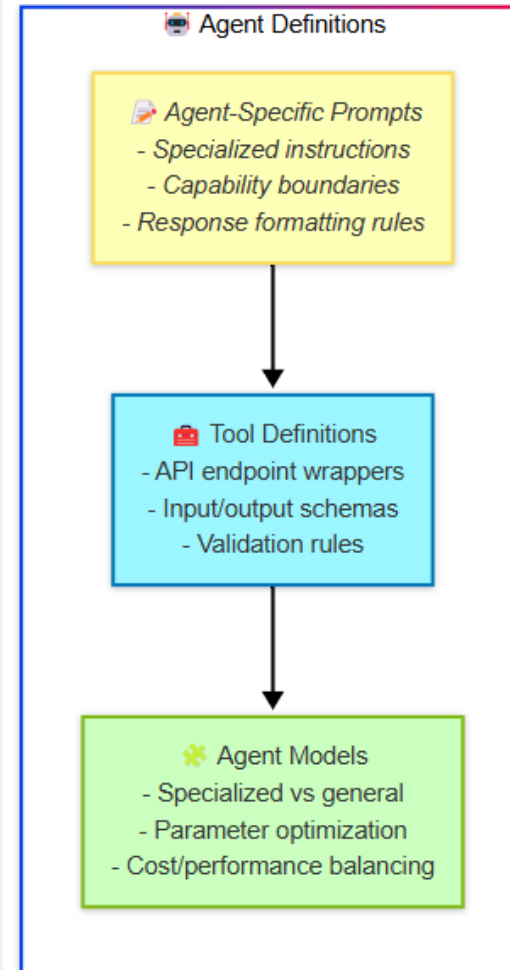
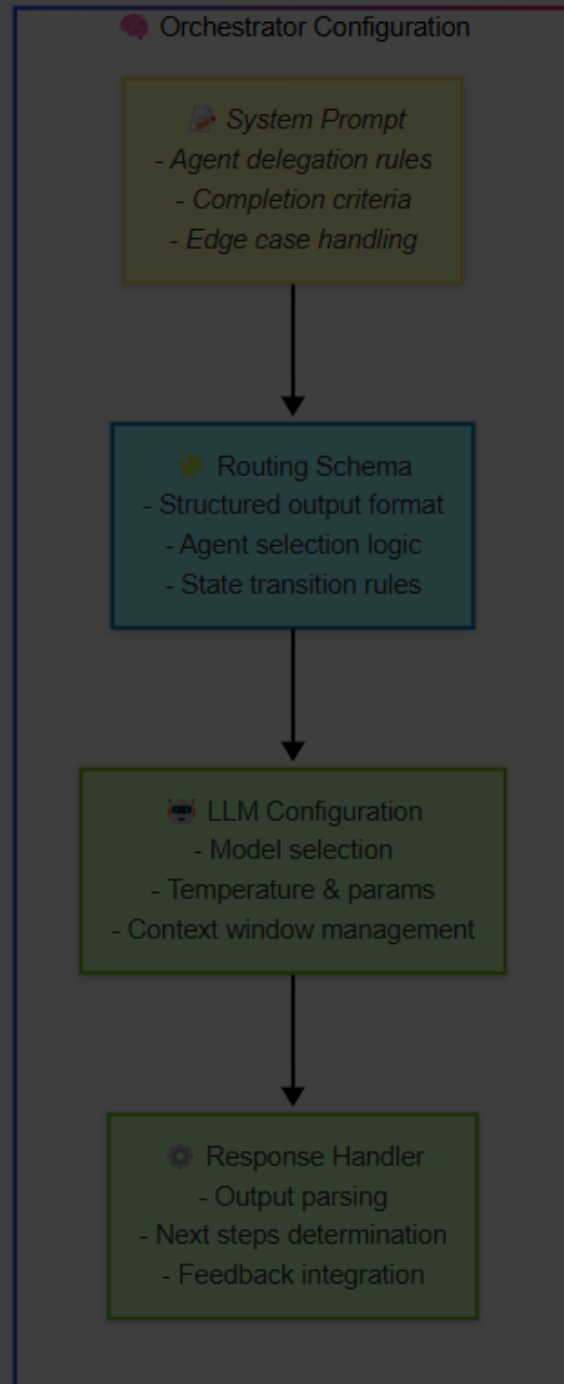
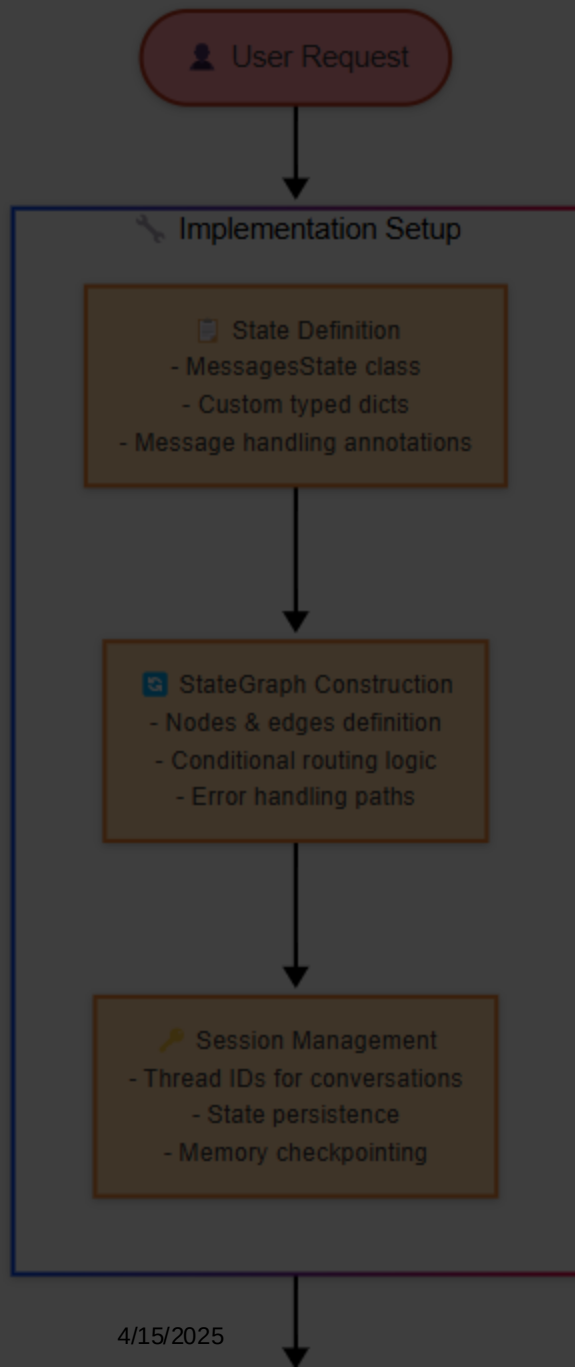
4/15/2025

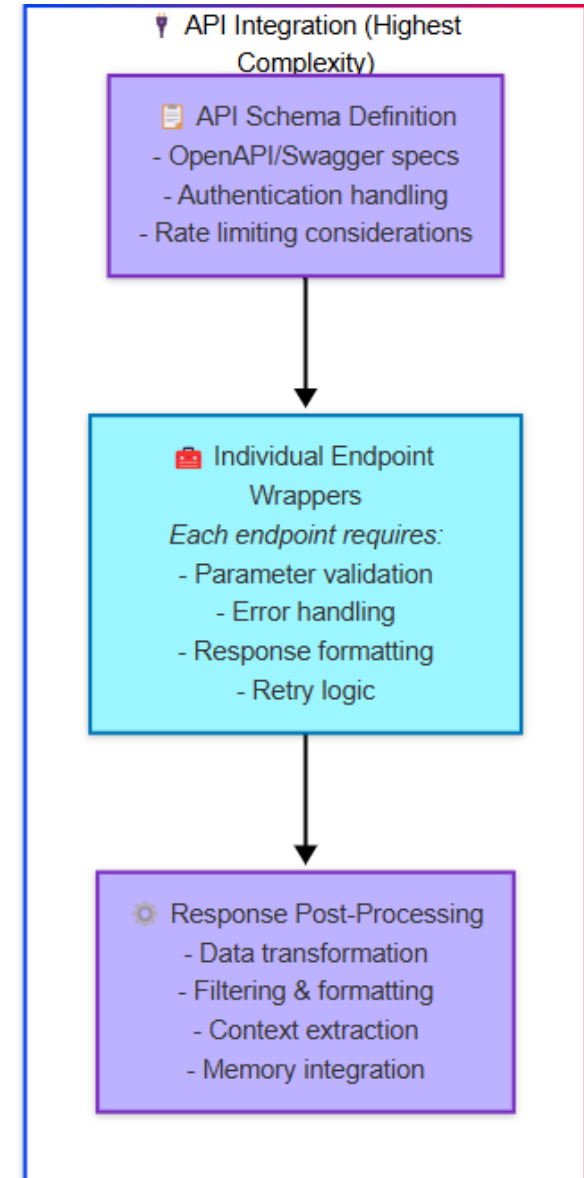
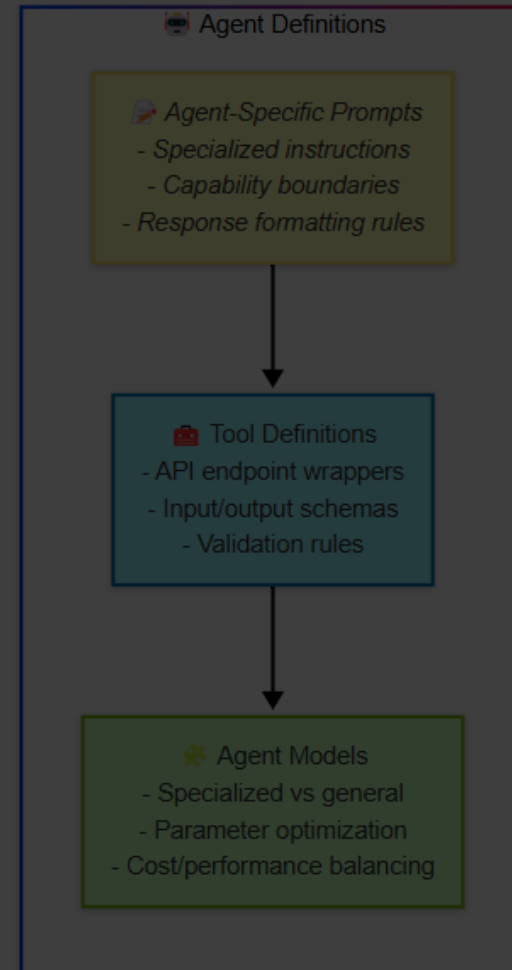
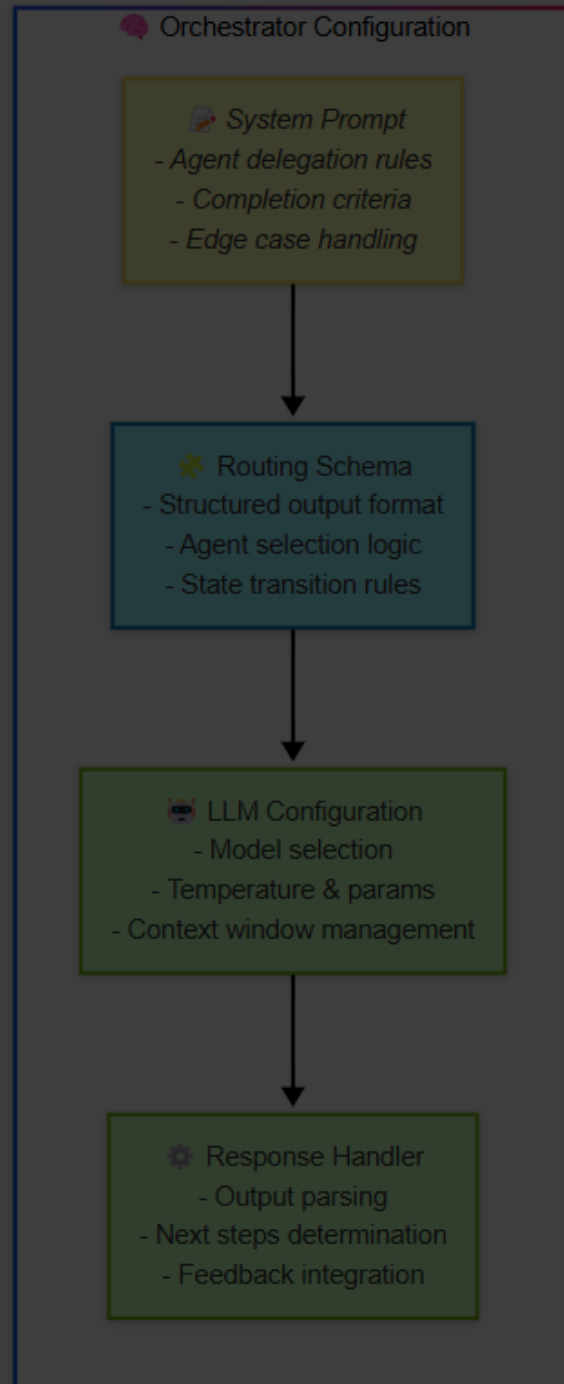
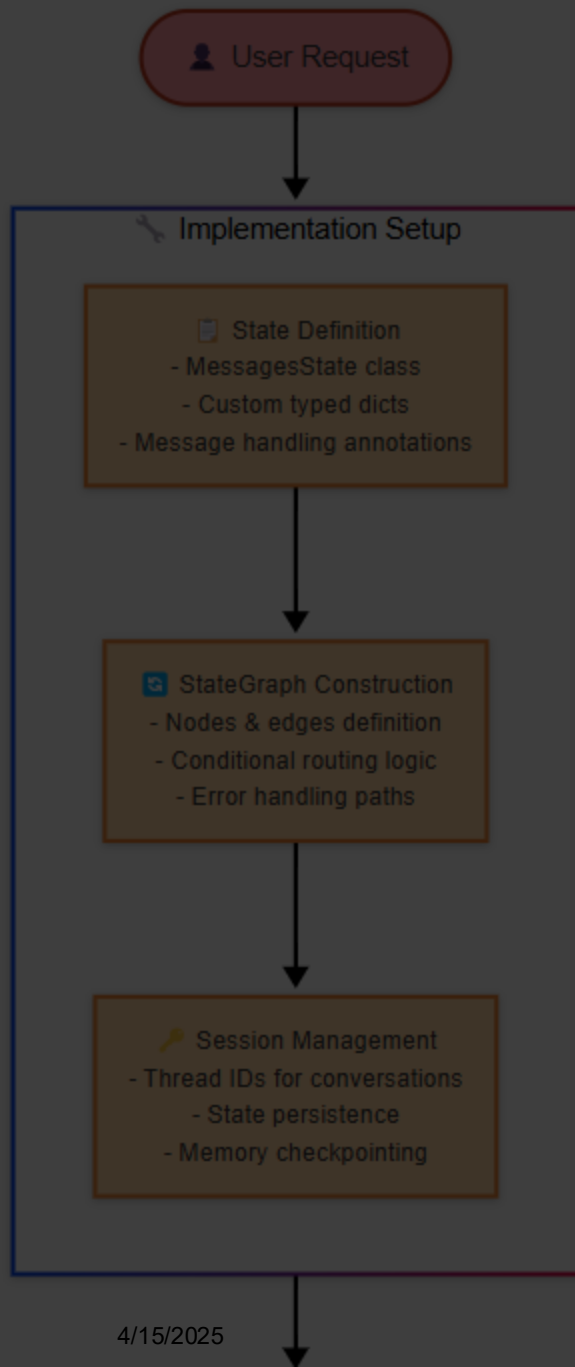












⚠ Key Engineering Challenges



Complex Prompt Engineering
Requires extensive testing & refinement



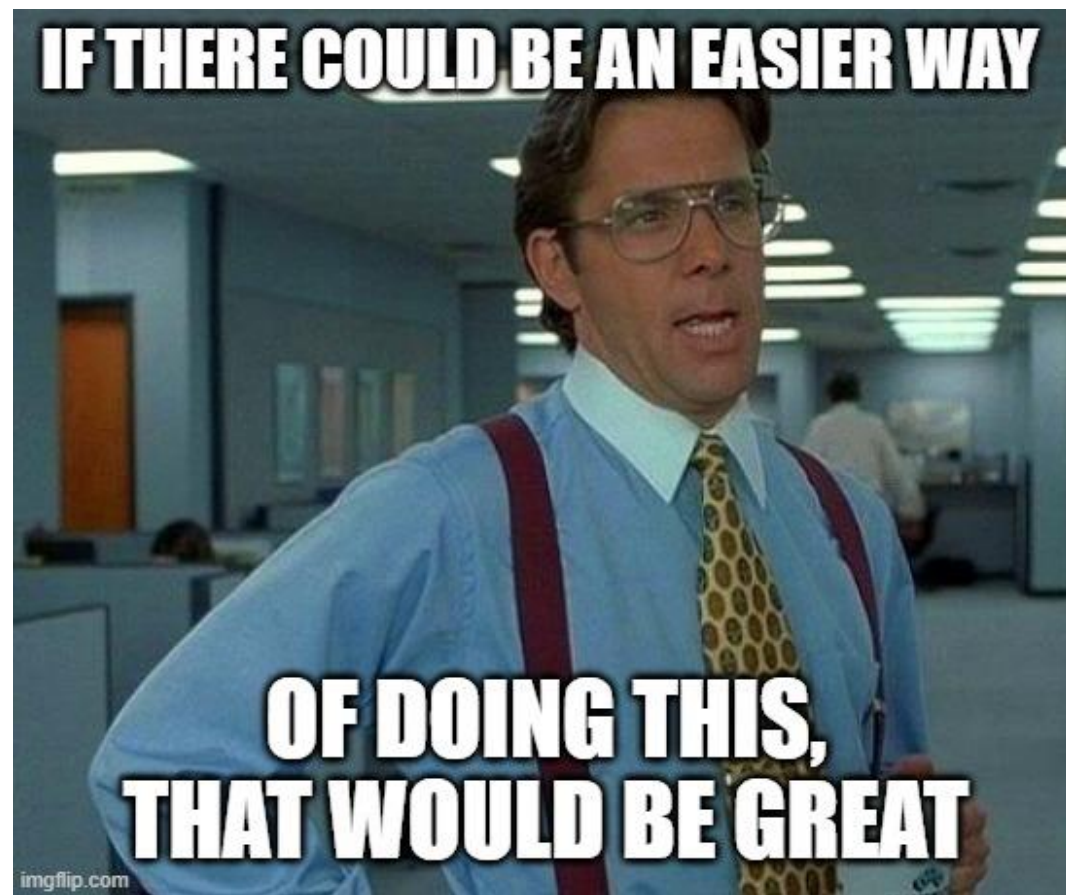
State Management Complexity
Thread safety & conversation tracking

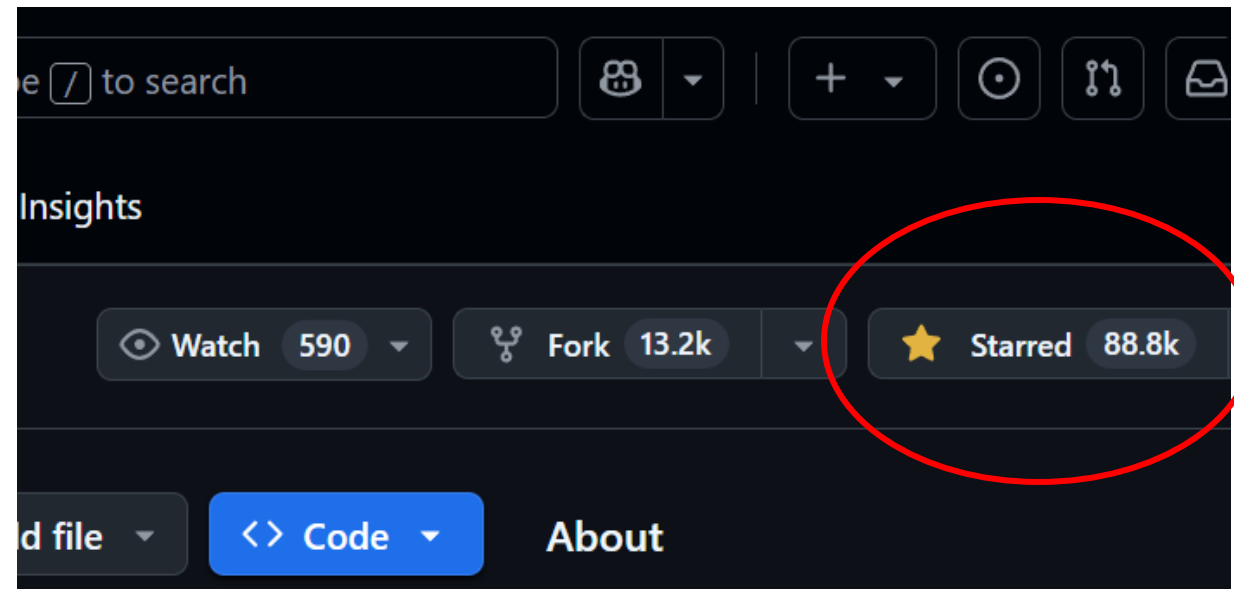
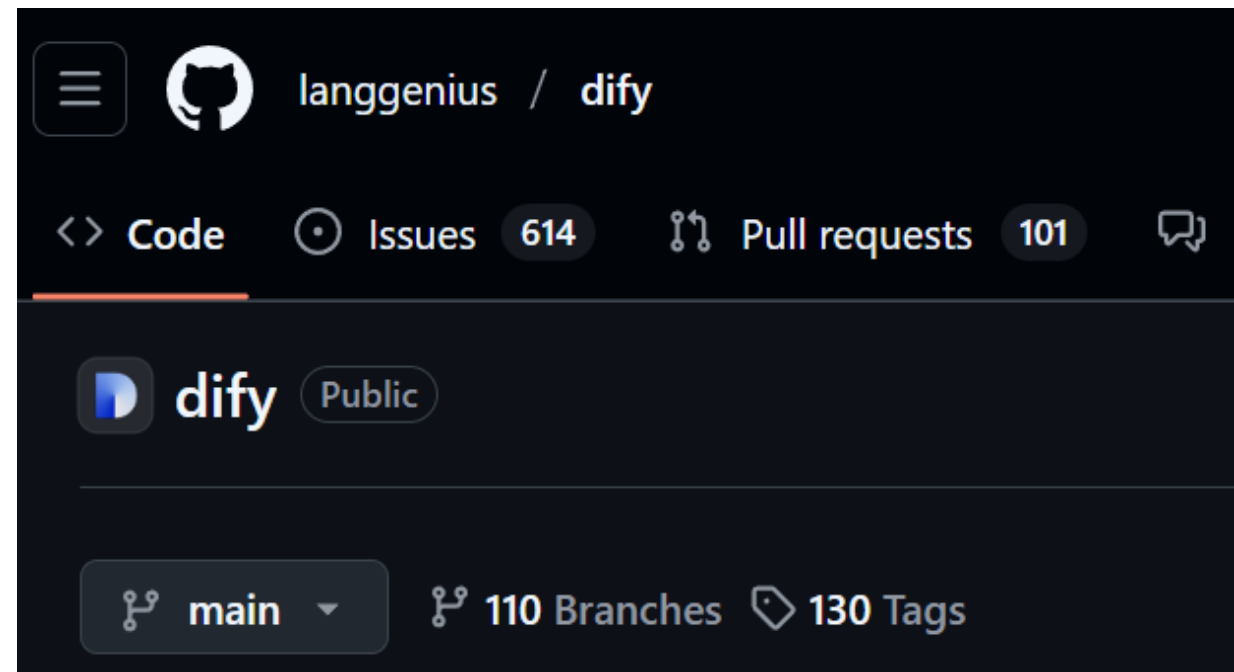


Tool Integration Overhead
Each API endpoint = custom tool wrapper



Testing & Observability
Difficult to validate agent behavior



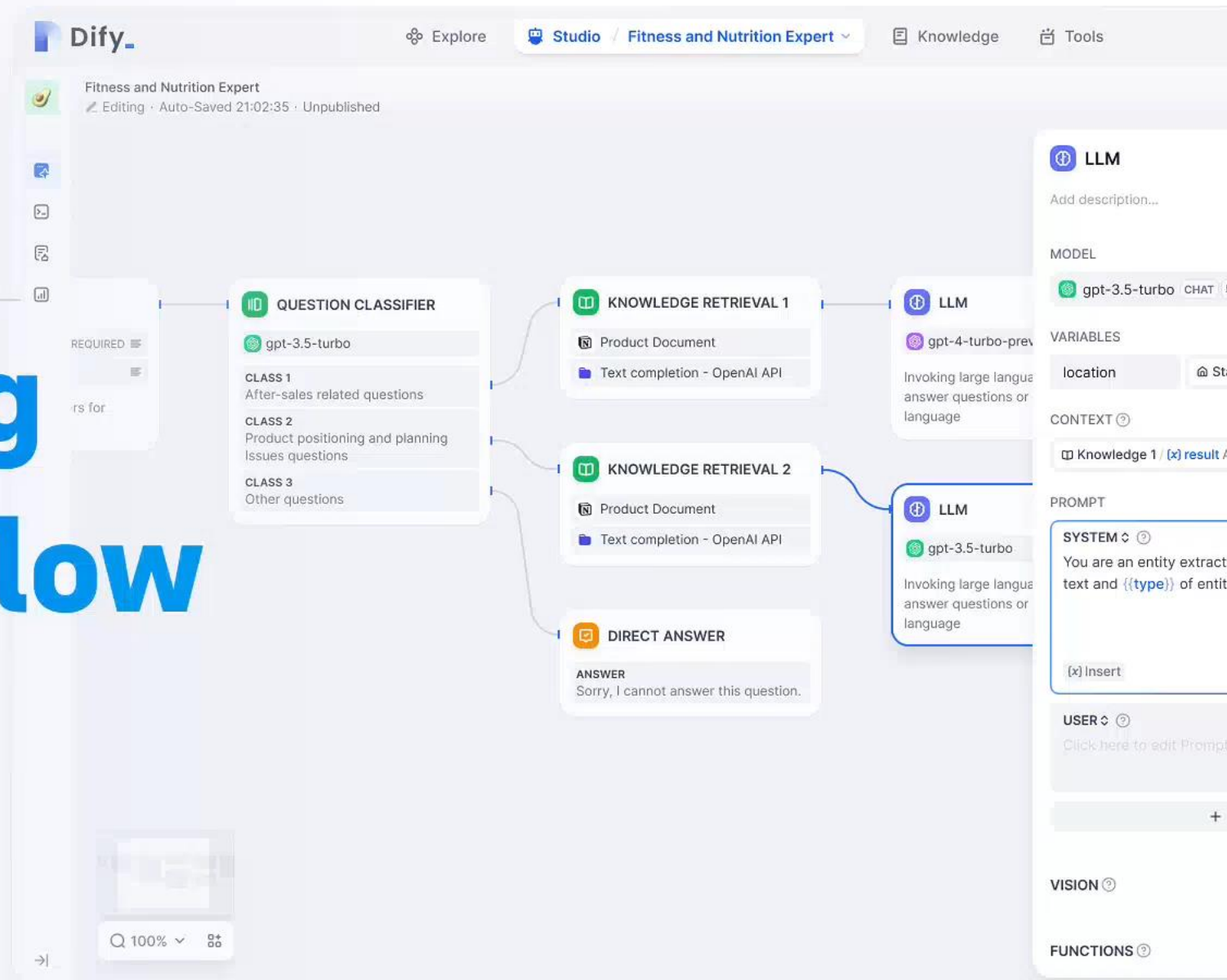


Feature Comparison

Feature	Dify.AI	LangChain	Flowise	OpenAI Assistants API
Programming Approach	API + App-oriented	Python Code	App-oriented	API-oriented
Supported LLMs	Rich Variety	Rich Variety	Rich Variety	OpenAI-only
RAG Engine	✓	✓	✓	✓
Agent	✓	✓	✗	✓
Workflow	✓	✗	✓	✗
Observability	✓	✓	✗	✗
Enterprise Feature (SSO/Access control)	✓	✗	✗	✗
Local Deployment	✓	✓	✓	✗



Introducing Dify Workflow



franzuzz's ...

ExploreStudio / Demo DifyKnowledgeTools

PLUGINSfranzuzz

Auto-Saved 11:48:55 · Published 42 minutes ago

START

QUESTION CLASSIFIER

LLM 4

LLM 2

GETCURRENTWEATHER

WEB SCRAPER

POST PROCESS

ANSWER

ANSWER 2

gpt-4o-mini

gpt-4o-mini

gpt-4o-mini

gpt-4o-mini

gpt-4o-mini

gpt-4o-mini

gpt-4o-mini

CLASS 1
WEATHER QUESTIONS

CLASS 2
Web Scraiping skills

USER_AGENT Mozilla/5.0 (Windows ...
GENERATE_SUM...

ANSWER
Post Pro... / (x) text

ANSWER
Post Pro... / (x) text

113%

React Flo

franzuzz's ...

Explore

Studio / basic_test_dify

Knowledge

Tools

PLUGINS

F franzuzz

API Server

http://localhost/v1

Status In Service

API Key

Advanced Chat App API

Chat applications support session persistence, allowing previous chat history to be used as context for responses. This can be applicable for chatbot, customer service AI, etc.

Base URL

Code

http://localhost/v1

Authentication

The Service API uses `API-Key` authentication. *Strongly recommend storing your API Key on the server-side, not shared or stored on the client-side, to avoid possible API-Key leakage that can lead to serious consequences.*

For all API requests, include your API Key in the `Authorization` HTTP Header, as shown below:

Code

Authorization: Bearer {API_KEY}

Contents

Send Chat Message

File Upload

Stop Generate

Message Feedback

Next Suggested Questions

Get Conversation History Messages

Get Conversations

Delete Conversation

Conversation Rename

Speech to Text

Text to Audio

Get Application Basic Information

Get Application Parameters Information

Get Application Meta Information



KNOWLEDGE

API ACCESS

☐ All Knowledge ?☐ All Tags ▾

Search

External Knowledge API



Create Knowledge

Import your own text data or write data in real-time via Webhook for LLM context enhancement.

Connect to an External Knowledge Base →

Add an External Knowledge API



Name *

Please enter

API Endpoint *

[Learn how to create an External Knowledge API](#)

Please enter

API Key *

Please enter

Cancel

Save

🔒 Your API Token will be encrypted and stored using [PKCS1_OAEP](#) technology.

External Knowledge API



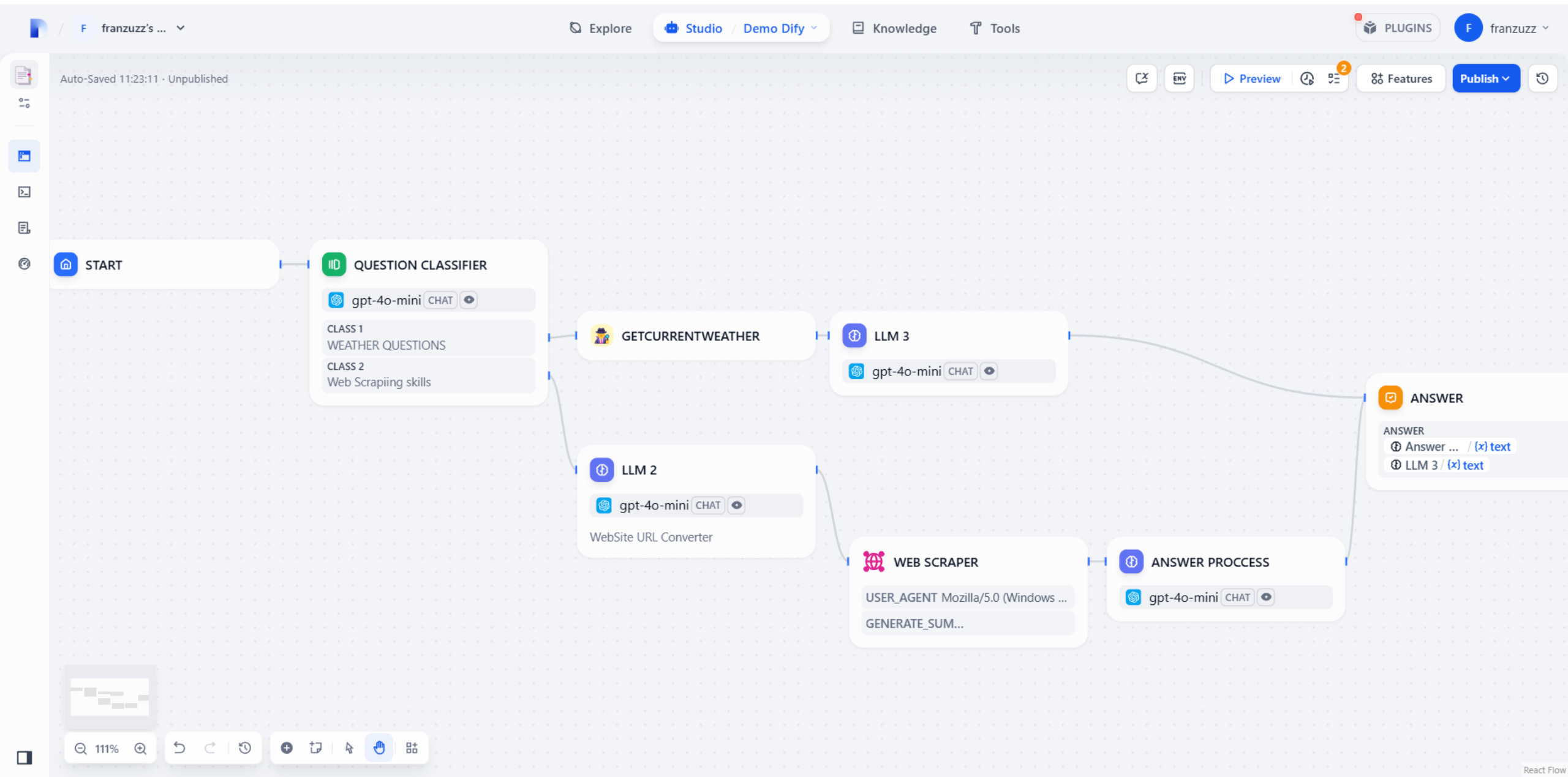
The external knowledge API is used to connect to a knowledge base outside of Dify and retrieve knowledge from that knowledge base.

[Learn how to create an External Knowledge API](#)

[+ Add an External Knowledge API](#)

Did you know?

The Knowledge can be integrated into the Dify application [as a context](#), or it [can be created](#) as a standalone ChatGPT index plug-in to publish



Settings

WORKSPACE

 Model Provider

 Members

 Data Source

 API Extension

GENERAL

 Language

Members

✕
ESC



franzuzz's Workspace

 Add

NAME

LAST ACTIVE

ROLES



franzuzz(You)
gutij...

a few seconds
ago

Owner

Add team member

✕

They can access your team data directly after signing in.

Email

Please input emails

Invited as Normal user

▼

✓ Normal

Only can use apps, can not build apps

Editor

Can build & edit apps

Admin

Can build apps & manage team settings

Settings

WORKSPACE

 Model Provider

 Members

 Data Source

 API Extension

GENERAL

 Language

Model Provider

Search

Models

 System Model Settings

 Azure OpenAI Service Model

LLM TEXT EMBEDDING SPEECH2TEXT TTS

Show Models >

 Add Model

To be configured

 Azure AI Studio

LLM RERANK

 Set up api-key or add model to use

 Add Model

 OpenAI

LLM TEXT EMBEDDING SPEECH2TEXT MODERATION TTS

 Set up api-key or add model to use

API-KEY

 Setup

 Add Model

Install model providers

Discover more in [Dify Marketplace](#)



Anthropic
langgenius · 18,124

Anthropic's powerful models.



Amazon Bedrock
langgenius · 3,173

The models of Amazon Bedrock



Cohere
langgenius · 4,300

Cohere



Gemini
langgenius · 23,154

Google's Gemini model.



Hugging Face Hub
langgenius · 6,144

Hugging Face Model



Replicate
langgenius · 599

Replicate

Demo Dify - Dify

localhost/app/b5887143-e361-4483-ad66-a21ea1009774/workflow

ExploreStudio / Demo DifyKnowledgeTools

PLUGINSfranzuzz

Auto-Saved 11:45:33 · Published a minute ago

PreviewFeaturesPublish

Demo Dify

Get API Key for weather from
https://weatherstack.com/documentation

franzuzz

START

QUESTION CLASSIFIER

gpt-4o-mini

CHAT

CLASS 1

WEATHER QUESTIONS

CLASS 2

Web Scraiping skills

LLM 4

gpt-4o-mini

CHAT

GETCURRENTWEATHER

POST PROCESS

gpt-4o-mini

CHAT

ANSWER

Post Pro...

(x) text

LLM 2

gpt-4o-mini

CHAT

WebSite URL Converter

WEB SCRAPER

USER_AGENT Mozilla/5.0 (Windows ...

GENERATE_SUM...

POST PROCCSS

gpt-4o-mini

CHAT

ANSWER 2

Post Pro...

(x) text

98%

UndoRedoRefresh

Zoom

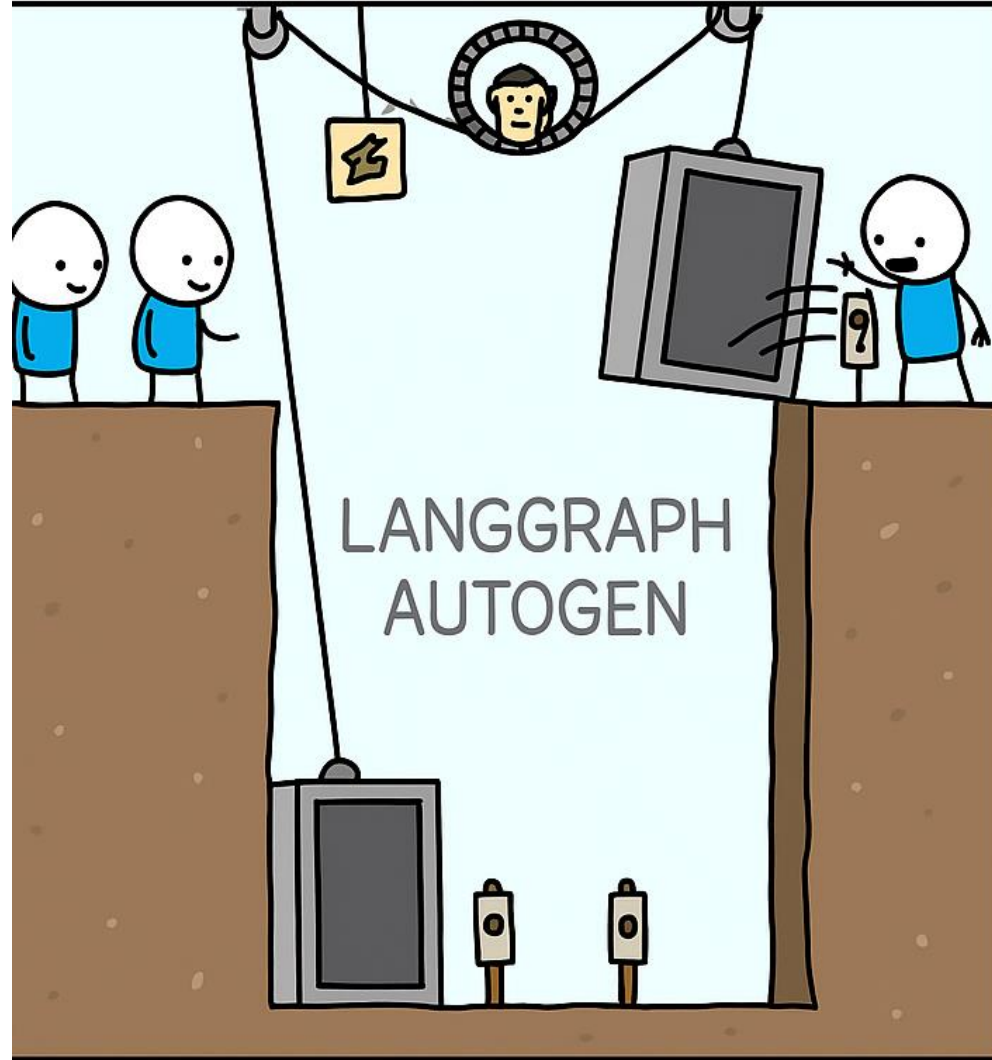
PREVIEW

Enter content in the box below to start debugging the Chatbot

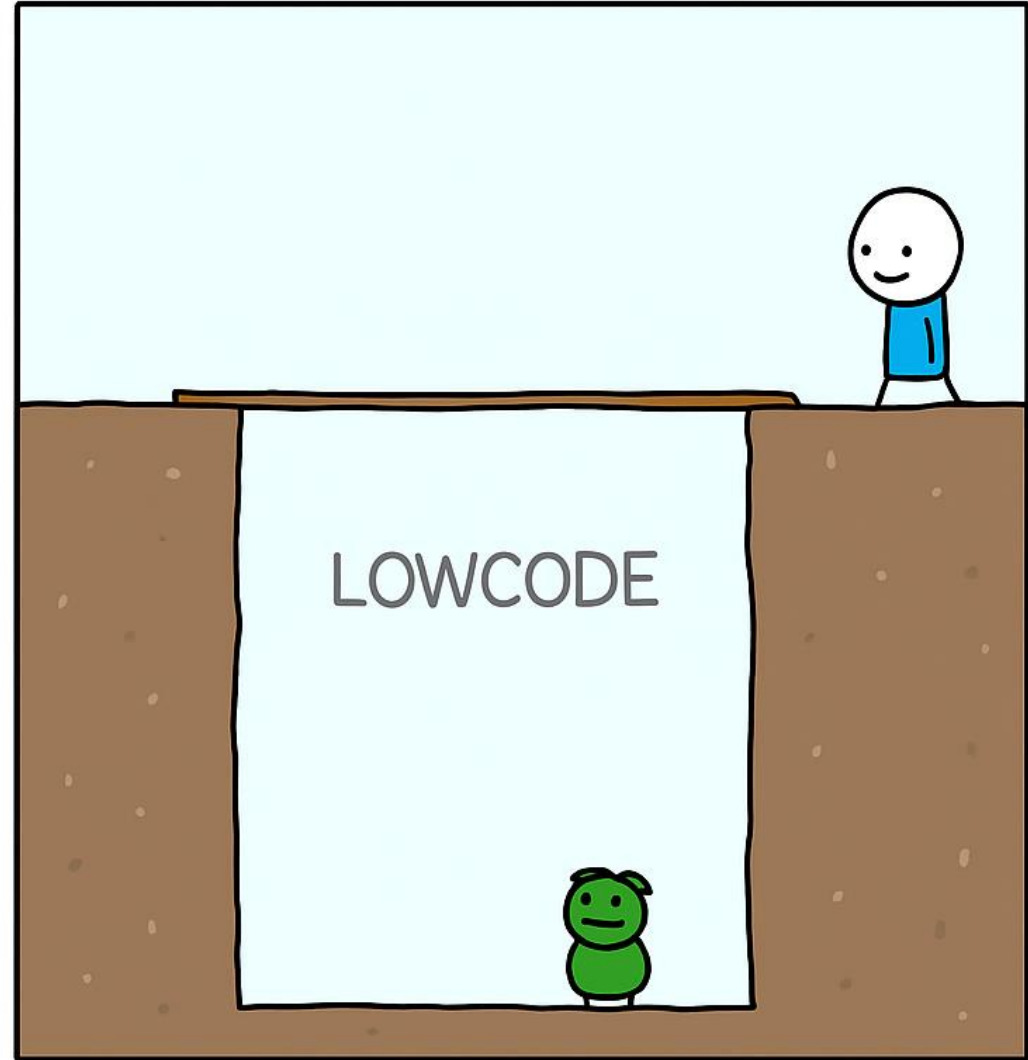
Talk to Bot

Enable feature to enhance web app user experience

COMPLEXITY



LANGGRAPH



LOWCODE



GitHub Copilot

Must have tips

- Agentic Mode
- System Prompt Rules & Docs
- MPC



Edit with Copilot

Agent Mode

Ask Copilot to edit your files in [agent mode](#). Copilot will automatically use multiple requests to pick files to edit, run terminal commands, and iterate on errors.

Copilot is powered by AI, so mistakes are possible. Review output carefully before use.

 6

 Add Context...

 .gitignore Current file 

Edit files in your workspace in agent mode



Agent ▾

- Claude 3.5 Sonnet (Preview)
- ✓ Claude 3.7 Sonnet (Preview)
- GPT-4o

Claude 3.7 Sonnet (Preview) ▾



EXPLORER

DEMO

demo.md

OUTLINE

TIMELINE

demo

Show All Commands

Ctrl + Shift + P

Go to File

Ctrl + P

Open Chat

Ctrl + Alt + I

Toggle Terminal

Ctrl + ñ

Find in Files

Ctrl + Shift + F

CHAT

Edit with Copilot

Agent Mode

Ask Copilot to edit your files in agent mode. Copilot will automatically use multiple requests to pick files to edit, run terminal commands, and iterate on errors.

Copilot is powered by AI, so mistakes are possible. Review output carefully before use.

or type # to attach context

@ to chat with extensions

Type / to use commands

6 Add Context...

Write a pygame of a mario bros game, but its called mario sisters. Its sarirical. Make it even better, the best you can. I dont want very large files so please try to be modular. Good luck Think Hard

Agent Claude 3.7 Sonnet (Preview)

📄

🔍

🔗

🔧

📦

🖥️

🧪

🌊

🗣️

☰

👤

⚙️

Welcome

Untitled-1

1

Press `Ctrl` + `I` to ask GitHub Copilot to do something. Start typing to dismiss.

🔍 Search

👤

CHAT

COPILOT EDITS

👤

Ask Copilot

Copilot is powered by AI, so mistakes are possible. Review output carefully before use.

📎

 or type # to attach context

@ to chat with extensions

Type / to use commands

/fix the problems in my code

/tests add unit tests for my code

/explain how the selected code works

📎 Add Context

Untitled-1 Current file

@mcp

 I need a example of Azure AI Agents from the Foundry being used with semantic kernel. Seach on perplexity first for updated docs.

@

🗣️

o3-mini (Preview)

Ln 1, Col 1

Spaces: 4

UTF-8

CRLF

{}

Python

3.12.6

🔔



Jean-François (Fran) Gutierrez ✓ (He/Him)

AI Augmented Software Engineering Advocate | 🤖 A.K.A. "The Go-To Guy for Gen AI Insights"

Valencia, Valencian Community, Spain · [Contact info](#)



🤖 Tailor-Made AI Solutions

💻 Software Consultancy

🎤 Tech Events

👉 Google Developer Groups



Formación



/GutierrezFrancois



GitHub

/DrZuzzjen



/DrZuzzjen



HUGGING FACE
/FranZuzz

